

SOLUTION BRIEF

Agentic Coding - Shared Memory Layer

One shared memory for a team of agents.

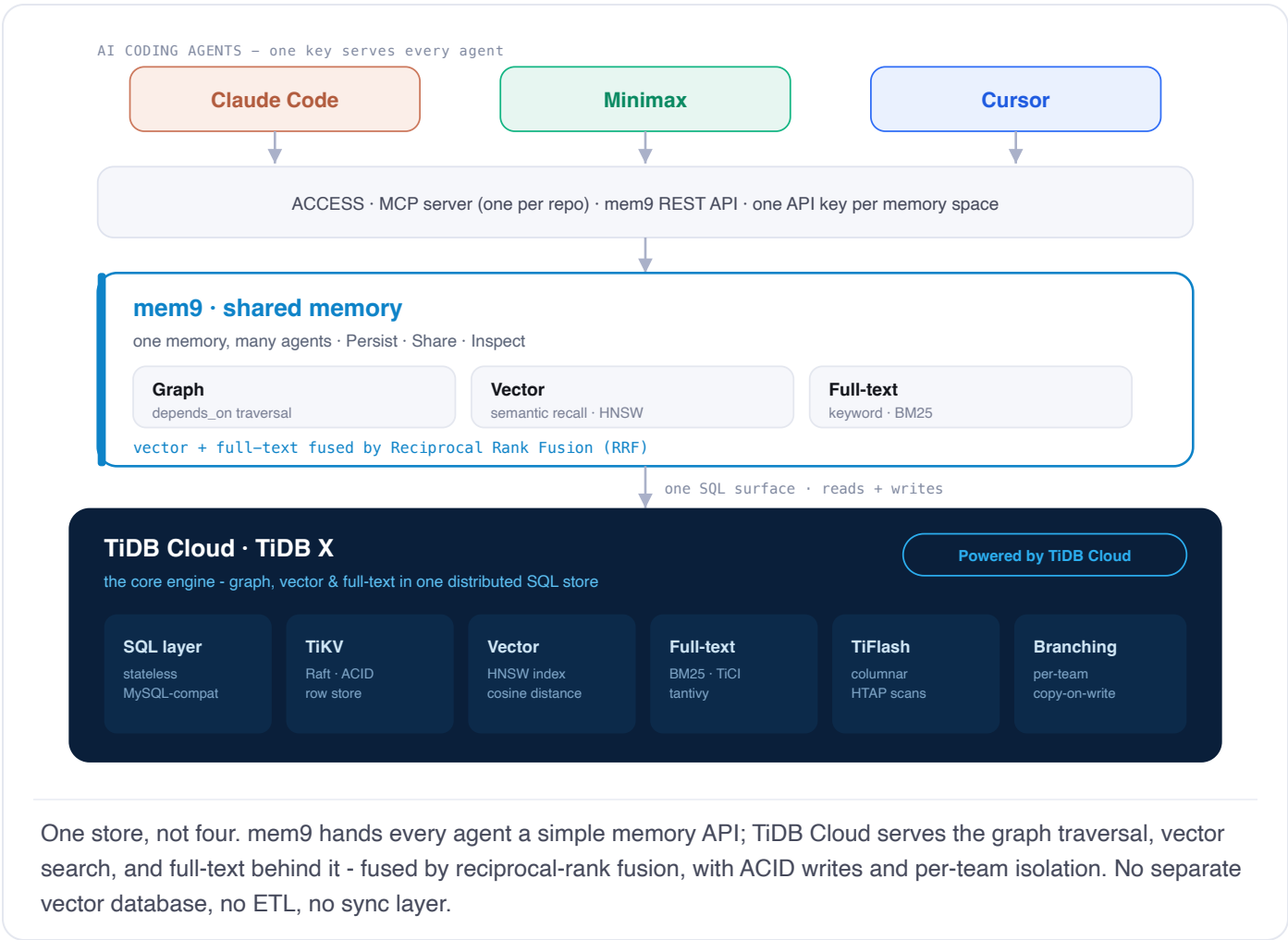
Bottom line. mem9 gives a team of AI coding agents one shared, queryable memory on TiDB Cloud. Agents read before they build and write back when done - so none starts cold or duplicates work. It is isolated per team, an agent can work across repos in a single session, and it answers the one question a vector store cannot: the transitive blast radius of a change. Every screenshot in this brief is a live read or write against TiDB Cloud.

THE PROBLEM	HOW MEM9 SOLVES IT	SHOWN IN
Each team's knowledge must stay private - no team reading another's.	One TiDB Cloud cluster per team (a mem9 space). A team's key reaches only its own memory; isolation is enforced by the platform, not application code.	§ The model
One integration should serve many repositories, not one wiring per repo.	Within a team, repos are <code>appId</code> namespaces on one cluster and the agent routes to the right one . Across teams, separate keys mean separate clusters.	§ The model
A single task often spans repos - create a resource in one, the dependent resource in another.	One agent session writes to each repo's namespace and links the records by <code>account_ref</code> - one integration, two repos, no manual hand-off.	§ One task, two repos
Some teams require a self-hosted deployment.	Supported. Vector and graph run self-hosted; full-text search is TiDB Cloud only today.	§ Deployment

Demonstrated live with three different coding agents – Claude Code, Minimax, and Cursor – against a single TiDB Cloud cluster. The tools are interchangeable; the shared memory is the point.

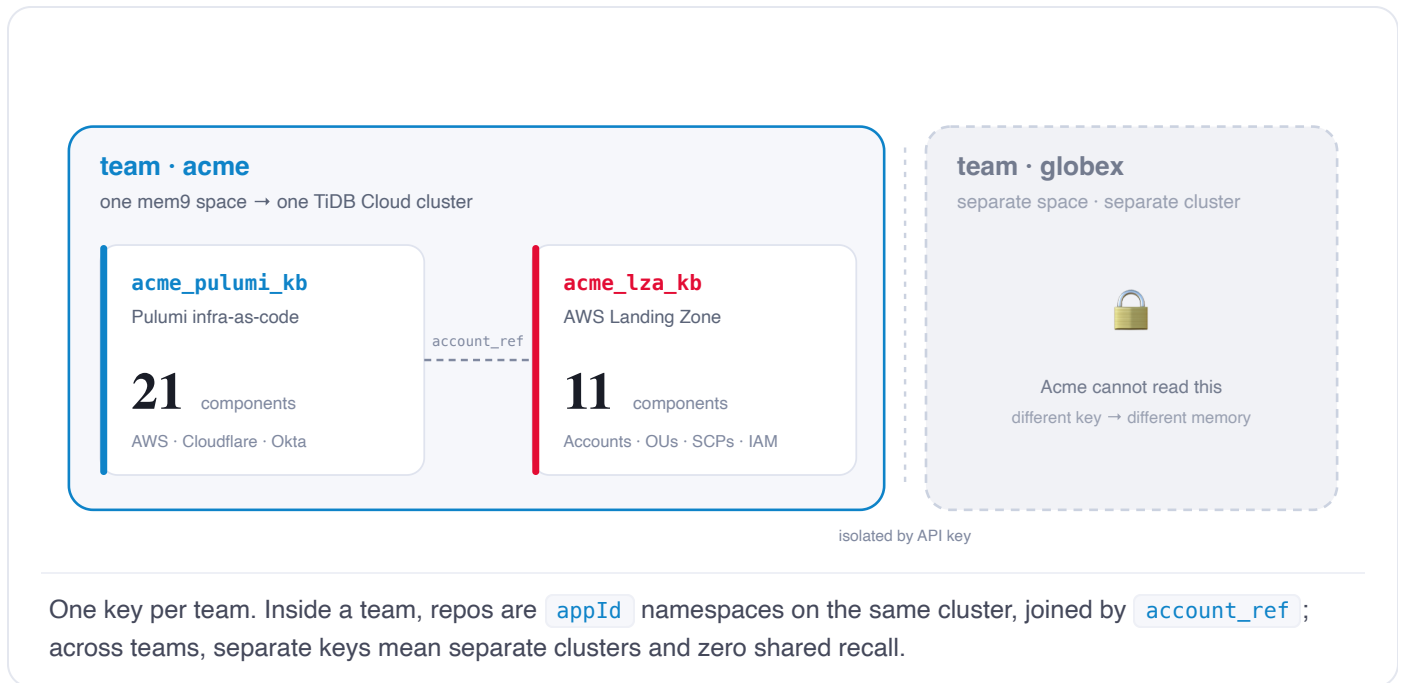
mem9 is the memory. TiDB is the engine.

mem9 is the agent-facing product - **one memory, many agents**: a shared, persistent layer every coding agent reads and writes through a single key, with hybrid recall and a human-inspectable dashboard. Underneath it, **TiDB Cloud is the core engine** - graph relationships, vector embeddings, and full-text indexes all live in one distributed SQL store. mem9's own site says it plainly: *"Powered by TiDB Cloud."*



Team = cluster. Repo = namespace.

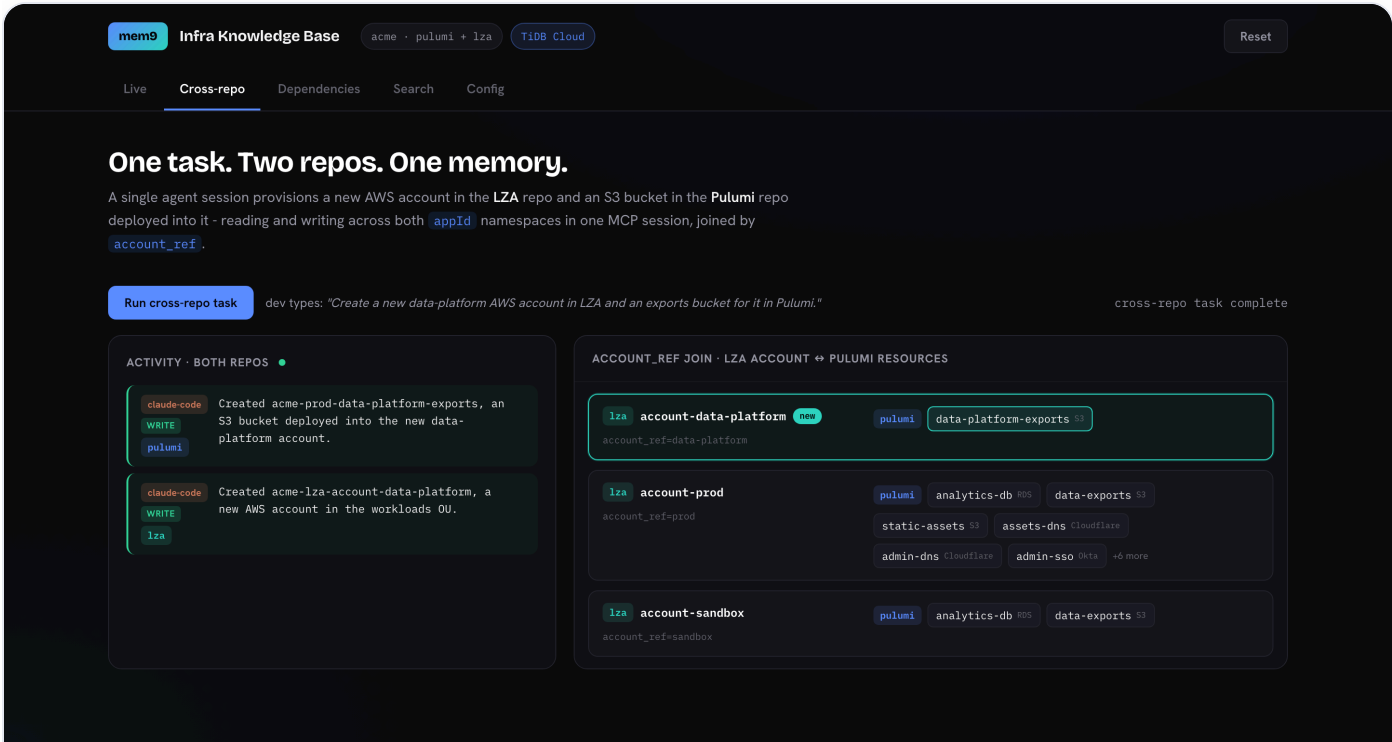
mem9 maps onto how a team already works. A **team** is a mem9 **space** - one key, one isolated TiDB Cloud cluster. A **repo** is an `appId` namespace inside it, so repos share a cluster but keep separate recall scopes. Another team is another cluster, reachable only with its own key.



Each team's space sits on its own managed TiDB Cloud cluster. Agents never handle raw database credentials – the mem9 API key is the only secret.

One task, two repos.

A single task often spans repos - create a new AWS account in the LZA repo *and* the S3 bucket for it in the Pulumi repo. One agent, given one plain-language request, writes the account to `acme_lza_kb` and the bucket to `acme_pulumi_kb` - two separate namespaces - and links them with `account_ref`. No second session, no manual hand-off.



DASHBOARD · CROSS-REPO Left: one activity feed shows the same agent writing to **both** repos. Right: the `account_ref` join - the new **data-platform** account and its Pulumi exports bucket, created together and linked.

Why it matters. The agent doesn't need to be told which repo owns what. It routes each write to the right namespace and records the cross-repo link, so the next session - in either repo - already knows the account and its bucket belong together.

Three tools, one memory, warm hand-offs.

Production is complete; staging has drifted by four components. Three different agents bring it back to parity - the developer only states the goal. Every agent's rules file ([CLAUDE.md](#) / [AGENTS.md](#) / [.cursorsrules](#)) makes the same first move non-negotiable: **query mem9 before writing anything**. So each step below runs the same shape - recall, compose, write back.

The screenshot displays the mem9 Infra Knowledge Base interface. At the top, there's a header with 'mem9 Infra Knowledge Base', a search bar containing 'acme - pulumi + lza', and a 'TiDB Cloud' button. Below the header, there's a navigation bar with 'Live', 'Cross-repo', 'Dependencies', 'Search', and 'Config'. The main content area is titled 'Three tools, one memory, live.' and includes a sub-header: 'Every query and write from Claude Code, Minimax, and Cursor lands in the same mem9 space. Watch the staging gap close as each tool reads the shared log and writes back.'

The interface is divided into two main sections. On the left, the 'ACTIVITY' section shows 'No activity yet. Launch a CLI and ask it to query the knowledge base.' On the right, the 'KNOWLEDGE GRAPH' section displays a network of nodes and edges. The nodes are labeled with icons and names: 'analytics-db', 'data-exports', 'static-assets', 'assets-dns', 'data-kms', 'ScpPolicy', 'portal-svc', and 'static-assets'. The edges represent dependencies between these components. Below the graph, there's a 'STAGING PARITY' section showing a list of components and their status in staging. The status is '4 missing in staging'.

Component	Status
analytics-db	in staging
data-exports	in staging
static-assets	missing
assets-dns	missing

At the bottom of the interface, there's a 'STARTING POINT' section with the text: 'Feed empty; the staging-parity strip shows 4 missing.'

STEP 1 Claude Code

dev types "Bring staging up to parity with production – start with the static-assets bucket and its CDN."

QUERY-FIRST – the rules file requires a recall before any write



recall the gap → compose the S3 + DNS libraries → write both components back, with the dns-fronts-bucket edge.

Result. Staging gains the **static-assets** bucket and its **CDN**. Gap 4 → 2.

mem9

Infra Knowledge Base

acme · pulumi · lza

TiDB Cloud

Reset

LiveCross-repoDependenciesSearchConfig

Three tools, one memory, live.

Every query and write from Claude Code, Minimax, and Cursor lands in the same mem9 space. Watch the staging gap close as each tool reads the shared log and writes back.

ACTIVITY

claude-code

WRITE

Created acme-staging-assets-dns, a Cloudflare DNS fronting acme-staging-static-assets.

claude-code

WRITE

The acme-staging-static-assets resource is an S3 bucket composing the S3Bucket library.

KNOWLEDGE GRAPH

AllLibProdStagingOrg

LibraryS3RDSCloudflareOktaAccount

2 missing in staging

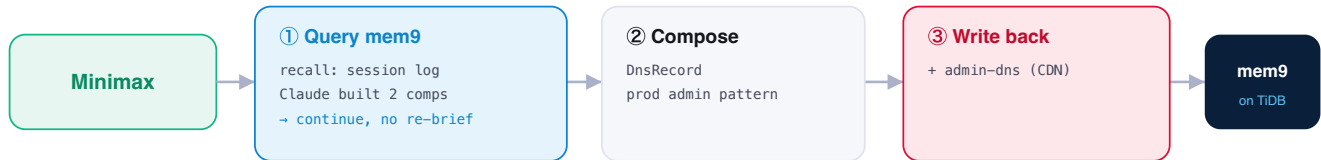
✓ analytics-db	RDS	In staging
✓ data-exports	S3	In staging
✓ static-assets	S3	In staging
✓ assets-dns	Cloudflare	In staging

RESULT · CLAUDE CODE Both writes land in the shared feed; the parity strip drops to **2 missing**.

STEP 2 Minimax

dev types "Keep bringing staging to parity with prod - it still needs the admin-portal DNS."

QUERY-FIRST - Minimax opens cold and reads the shared log before it builds



recall the shared session log (what Claude just built) → compose the admin DnsRecord → write it back. No re-briefing.

Result. The admin-portal **DNS** lands - Minimax never had to be re-briefed. **Gap 2 → 1.**

mem9

Infra Knowledge Base

acme · pulumi + lza

TiDB Cloud

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Live

Cross-repo

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Three tools, one memory, live.

Every query and write from Claude Code, Minimax, and Cursor lands in the same mem9 space. Watch the staging gap close as each tool reads the shared log and writes back.

ACTIVITY

minimax

WRITE

Created acme-staging-admin-dns, a Cloudflare DNS matching the production admin pattern.

minimax

QUERY

The user is continuing staging work.

claude-code

WRITE

Created acme-staging-assets-dns, a Cloudflare DNS fronting acme-staging-static-assets.

claude-code

WRITE

The acme-staging-static-assets resource is an S3 bucket composing the S3Bucket library.

KNOWLEDGE GRAPH

All Lib Prod Staging Org

Library S3 RDS Cloudflare Okta Account

pulumi lza

STAGING PARITY

1 missing in staging

✓ analytics-db	RDS	in staging
✓ data-exports	S3	in staging
✓ static-assets	S3	in staging
✓ assets-dns	Cloudflare	in staging

RESULT · MINIMAX Its query ("continuing staging work") and write join the same timeline as Claude Code's. **1 missing.**

STEP 3

Cursor

dev types "Finish bringing staging to parity – add the admin SSO app."

QUERY-FIRST – Cursor traces the dependency chain before it builds



hybrid-search the prod SSO app → trace its depends_on / redirects_to chain → compose and write the matching staging SSO app.

Result. The SSO app's redirect URI points at the admin DNS; staging reaches **parity**. **Gap 1 → 0.**

mem9

Infra Knowledge Base

acme · pulumi + lza

TiDB Cloud

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Three tools, one memory, live.

Every query and write from Claude Code, Minimax, and Cursor lands in the same mem9 space. Watch the staging gap close as each tool reads the shared log and writes back.

ACTIVITY

cursor

WRITE

Created acme-staging-admin-ss0, an Okta SSO app whose redirect URI points at acme-staging-admin-dns.

cursor

WRITE

Staging is now at parity with production.

minimax

WRITE

Created acme-staging-admin-dns, a Cloudflare DNS matching the production admin pattern.

minimax

QUERY

The user is continuing staging work.

claude-code

WRITE

Created acme-staging-assets-dns, a Cloudflare DNS fronting acme-staging-static-assets.

claude-code

WRITE

The acme-staging-static-assets resource is an S3 bucket composing the S3Bucket library.

KNOWLEDGE GRAPH

AllLibProdStagingOrg

LibraryS3RDSCloudflareOktaAccount

STAGING PARITY

at parity ✓

✓

analytics-db

RDS

In staging

✓

data-exports

S3

In staging

✓

static-assets

S3

In staging

✓

assets-dns

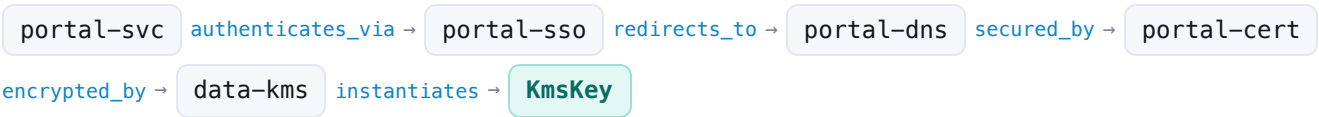
Cloudflare

In staging

RESULT · CURSOR One timeline now spans all three tools; the final write reads "Staging is now at parity with production." **0 missing.**

The question similarity can't answer.

A vector store finds things that *look* related. It cannot compute a transitive closure - "what, exactly, breaks if I change this?" That is graph reachability. mem9 records every `depends_on` and composition edge; one recursive traversal walks the chain to any depth.



Run that in reverse from `KmsKey` - "what breaks if we rotate it?" - and the answer is **13 components across four depths**. No flat search surfaces this; only walking the graph does.

mem9

Infra Knowledge Base

acme · pulumi + 12a

TiDB Cloud

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Transitive dependencies, in one pass.

Recursive reachability over the dependency graph - the question a vector store can't answer. Pick a component: trace what it **needs**, or what **breaks** if it changes. Each hop is a column.

Blast radius

Dependencies

KmsKey

TRY

rotate KmsKey →

portal-svc needs →

account-prod →

13 components · 4 hops · upstream

root

KmsKey

library

HOP 1

2

d1

data-kms

production

HOP 2

4

d2

analytics-db

production

HOP 3

4

d3

reporting-svc

production

HOP 4

3

d4

admin-sso

production

d1

TlsCertificate

library

d2

data-exports

production

d2

static-assets

production

d2

portal-cert

production

d3

assets-dns

production

d3

admin-dns

production

d3

portal-dns

production

d4

portal-sso

production

d4

portal-svc

production

LONGEST CHAINS

5h KmsKey - encrypted_by - TlsCertificate - instantiates - portal-cert - secured_by - portal-dns - redirects_to - portal-sso - uses - portal-svc

5h KmsKey - encrypted_by - TlsCertificate - instantiates - portal-cert - secured_by - portal-dns - redirects_to - portal-sso - authenticates_via - reporting-svc

5h KmsKey - instantiates - data-kms - encrypted by - portal-cert - secured by -

HOW IT'S RESOLVED

Blast radius of 'KmsKey': 13 components across 4 hop(s), transitively (KmsKey is needed by ...). Graph reachability over mem9 (appId=acme_pulumi_kb_demo).

DASHBOARD · DEPENDENCIES "What breaks if we rotate **KmsKey**?" Each hop is a column; 13 components fan across four depths and funnel onto the anchored foundation. The Trace panel reads each chain as a sentence.

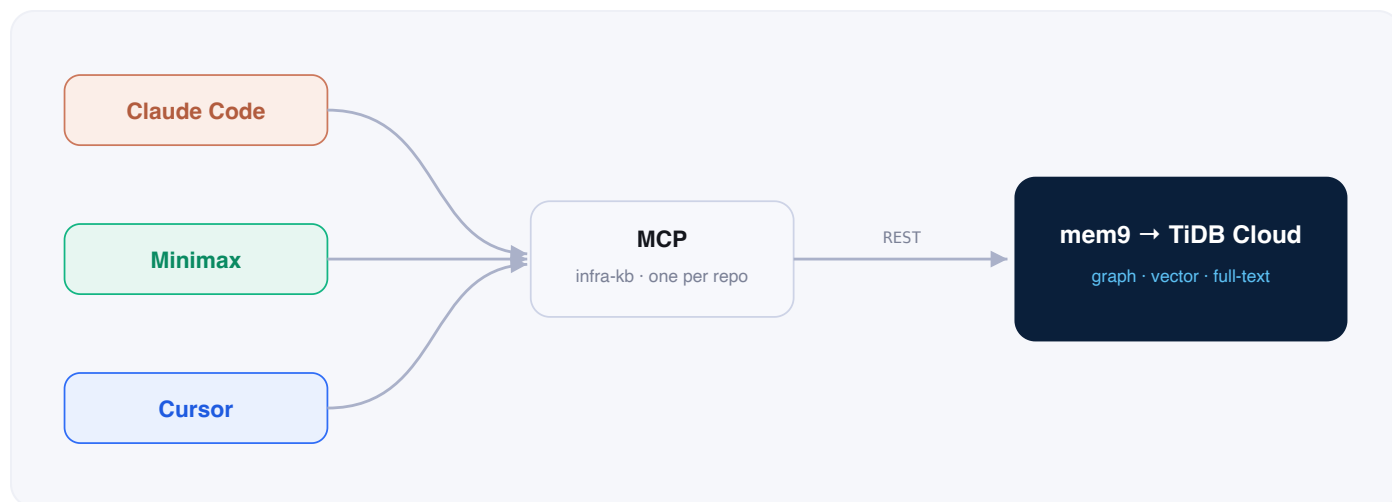
One engine, not four

APPROACH	GRAPH REACHABILITY	SEMANTIC RECALL	ATOMIC AGENT STATE
S3 / filesystem	no traversal	no	no transactions
Vector DB only	similarity ≠ reachability	yes	no
Four separate systems	partial, with ETL	yes	no cross-system atomicity
mem9 → TiDB Cloud	recursive traversal	vector + full-text	one transaction

Graph, vector embeddings, and full-text live in the same store - the same recall that answers "*what encrypts production data*" also computes the blast radius above. No separate vector DB, no ETL, consistent across concurrent writes from all three tools.

S3 is the filing cabinet for artifacts. mem9 is the brain agents query.

One store. One plug. One key.



MCP is the plug

Every tool loads the same Model Context Protocol config - one named server per repo. No custom glue per tool.

One store, every pattern

Relational state, the dependency graph, vector embeddings, and full-text all live in the same memory.

Just an API key

mem9 manages the TiDB Cloud cluster. Agents read and write through one key and never touch credentials.

The standing instruction

The developer types a normal request. The prescriptive part lives in each tool's rules file (`CLAUDE.md` / `AGENTS.md` / `.cursorrules`), so the agent queries first, composes, and writes back on every run with no per-task hand-holding.

mem9

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Behavior is configuration, not code.

The developer just types a normal request - "bring staging to parity." This standing instruction, loaded into every tool's rules file, makes the agent query first, compose, and write back on its own. No per-task hand-holding.

STANDING INSTRUCTION · CLAUDE.MD / AGENTS.MD / .CURSORRULES

Before you write or change any infrastructure - automatically, every task:

1. Query first. query_knowledge_base on the infra-kb MCP for the repo you're touching: what exists, how it connects, recent log.
2. Compose. Build from libs/ (S3Bucket, PostgresDatabase, DnsRecord, SsoApplication, KmsKey, ...). Never raw aws.*/cloudflare.*/okta.*.
3. Conventions. acme-<env>-<name> naming; required tags; DnsRecord proxied; SSO redirect -> DnsRecord; set account_ref (prod/sandbox).
4. Blast radius. Before changing a shared component, trace what depends on it.
5. Write back. After any change, call write_component to record it - or the next tool, and the next session, start cold.

The human never says these steps. They say what they want; you do this on your own.

MCP SERVERS · ONE ENTRY PER REPO

```
{
  "mcpServers": {
    "infra-kb-pulumi": { "command": "python", "args": ["-m", "src.mcp_server"] },
    "infra-kb-lza": { "command": "python", "args": ["-m", "src.mcp_server"] }
  }
}
```

query_knowledge_base · write_component -> hybrid recall + atomic writes

DASHBOARD · CONFIG

The five-step standing instruction every tool loads, and the one-entry-per-repo MCP wiring that makes it automatic.

Deployment

- **TiDB Cloud (shown here):** graph + vector + full-text, fully managed; nothing to host.
- **Self-hosted:** available - vector and graph run on-prem. Full-text search is TiDB Cloud only today, so this brief uses Cloud to show hybrid recall end to end.

mem9 · TiDB Cloud – shared memory for AI coding agents · v3. Live demo: Claude Code · Minimax · Cursor over team acme (repos pulumi + lza). Every figure is a live read/write against TiDB Cloud.